

Karthik Mukkera

Full Stack Developer

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Summary

Full Stack Developer with 5+ years of experience driving scalable, secure, and high-performance applications using Java, .NET Core, React, Angular, and cloud platforms (AWS, Azure). Expert in designing efficient databases with PostgreSQL, MongoDB, and SQL Server, and delivering HIPAA-compliant microservices that improve scalability and reduce latency. Led cloud migrations and automated CI/CD pipelines, achieving near-perfect uptime and accelerating deployment cycles. Enhanced user engagement through intuitive front-end development and strengthened code quality with comprehensive testing using Selenium, Jest, Cucumber, and Appium. Proven ability to align technology solutions with business goals and healthcare compliance, consistently reducing defects and boosting system reliability.

Skills

Languages: Java, JavaScript, Python, C#, TypeScript, SQL

Frameworks and Libraries: React.js, Next.js, Angular, HTML5, CSS3, Bootstrap, Node.js, Spring Boot, Hibernate, Django, Flask, .NET Core, RESTful APIs, JSON, TensorFlow

Databases: PostgreSQL, MySQL, MongoDB, Microsoft SQL Server

Cloud & DevOps: AWS (EC2, S3, Lambda), Azure, Docker, Kubernetes, Jenkins, Git, GitHub, Azure DevOps

Testing & Quality Assurance Tools: Selenium, Jest, Cypress, Appium, Katalon, Cucumber (BDD), JUnit, Postman, Kafka, Jira / Confluence

IDE: Eclipse, Visual Code, Notepad++, NetBeans, Dream Viewer

Architecture: Microservices, REST APIs, Serverless, CI/CD pipelines, Scalable Inference Pipelines, Real-Time Systems

Methodologies: Agile, Scrum, SDLC

Experience

Full Stack Developer | CVS Health

Jan 2024 - Present

- Led scalable, secure applications with React, Angular, TypeScript, and Spring Boot, boosting clinician and patient portal engagement by 60% and improving usability.
- Architected HIPAA-compliant RESTful APIs and microservices with Java, .NET Core, Spring Security, and Kubernetes, boosting backend scalability by 40% for real-time processing of billions of data points.
- Managed cloud deployments on Microsoft Azure and AWS, migrating legacy platforms to use EC2, S3, and Lambda, boosting system performance by 45% while optimizing uptime and scalability for mission-critical applications.
- Reduced production defects and bugs by 40% through comprehensive testing with Selenium, Appium, and Cucumber, ensuring code reliability.
- Automated CI/CD pipelines with Jenkins and Azure DevOps, containerized applications using Docker and Kubernetes, achieving 99.99% uptime, reducing deployment failures by 20%, and shortening deployment cycles.
- Conducted SQL and Python data analysis using Django and Flask to support clinical decisions and regulatory reporting, while implementing cloud monitoring to maintain 99.9% application availability.
- Designed efficient PostgreSQL and MongoDB schemas to improve query performance by 30%, collaborating with Agile teams to deliver features aligned with healthcare regulations and business goals.

Software Engineer | CueTech Systems

Jun 2019 - Jul 2022

- Designed, developed, and maintained scalable client- and server-side applications using JavaScript, React, Next.js, C#, .NET Core, and TensorFlow, improving user experience and accelerating feature delivery.
- Reduced infrastructure costs by 40% and boosted uptime by migrating monoliths to AWS cloud-native architecture; led microservices migration with Kafka, reducing latency by 25% and enabling real-time insights.
- Increased deployment frequency from weekly to daily by building automated CI/CD pipelines with Jenkins, GitHub Actions, and Azure DevOps, accelerating releases and reducing production issues.
- Improved system reliability by reducing incident resolution time through root cause analysis and proactive alerting on AWS cloud services, significantly cutting downtime.
- Boosted system performance by 40% and reduced churn by optimizing backend APIs and database queries in SQL Server, PostgreSQL, and NoSQL, enhancing scalability and data responsiveness.

- Applied software design patterns, microservices architecture, and automated testing with JUnit, Mockito, Jest, and Cypress to enhance code quality and maintainability.
- Led projects from concept to delivery by translating business needs into reusable code, cutting development time by 20%, while managing timelines and stakeholder expectations with Jira and Confluence for on-time delivery.

Education

Master of Science in Management Information Systems | *Lamar University*

Certificates

Achievement from AWS Web Services (AWS) - Solutions Architect Associate